PHAN NGUYEN MANH CUONG

Email: cuong040205a7@gmail.com Phone: 0393 403 012

GitHub: github.com/cuong4205

SUMMARY

Computer Science student interested in Backend Engineering, with hands-on experience building personal backend projects using NestJS, PostgreSQL, MongoDB, Redis and Docker. Eager to learn backend technologies, scalable system design and distributed systems concepts while continuously improving practical engineering skills.

EDUCATION

Hanoi University Of Science

Bachelor of Computer Science

Expected Graduation: 2027

GPA: 3.1/4

TECHNICAL SKILLS

- **Languages:** JavaScript, TypeScript, Java
- **Backend:** NestJS, Spring Boot
- **Databases:** PostgreSQL, MongoDB
- **Familiar:** Distributed systems concepts, Redis, Kafka, gRPC
- **DevOps:** Docker, Docker Compose, GitHub Action (CI/CD)
- **Tools:** Git, Linux, Postman

PROJECTS

Video Sharing Platform *Personal Project*

GitHub: github.com/cuong4205/Video-platform-Back-end

Overview

A microservice-based system using NestJS with separated services for learning distributed architecture concepts.

Key Features

- Separated upload, video processing and notification logic into independent services to explore service communication patterns.
- Built chunked upload mechanism for video files.
- Used Kafka for asynchronous workflows such as notifications and video-related events.
- Applied Redis caching for frequently accessed data to reduce repeated database queries.
- Containerized services using Docker and experimented with GitHub Actions for CI workflows.

Realtime Chat Application *Personal Project*

GitHub: github.com/cuong4205/Chat-app

Overview

Realtime chat application supporting direct messaging and group conversations using Socket.IO and Redis.

Key Features

- Experimented with real-time messaging using Socket.IO.
- Designed chat services following Microservices Architecture.
- Implemented asynchronous communication between services using Kafka.
- Using Redis TTL caching for frequently accessed data in backend services.
- Configured Nginx as an API gateway with per-route rate limiting.
- Implemented JWT Authentication with Refresh Token rotation.

ADDITIONAL INFORMATION

- Understanding of Object-Oriented Programming and Data Structures.
- Familiar with Distributed Systems and Event-Driven Architecture.
- Comfortable reading technical documentation in English.
- Interested in Backend Engineering, Cloud Infrastructure and System Design.